

What is Computer Science?

Week 2/16

1

Objectives -

At the end of this lecture, you should:

1. Have a wider perspective of what is Computer Science?
2. Be able to differentiate computer science from IT and Computer Engineering
3. Understand the relationship between Mathematics and Engineering

Introduction video:

[Should I go into Computer Science?](#)
[Is computer programming hard to learn?](#)

9/2/25

2

Outline of Topics

Nature of Computer Science

1. What is Computer Science?
2. What are the computer science areas?
3. Comparison between Computer science with other disciplines

Objectives - At the end of this session, you will be able to:

1. Understand what is computer science and its different areas of specializations;

3

What is Computer?

- **Computer** was derived from the French word **computer** or Latin word **computare**, which means '*to count, sum up; reckon*'.
- Computer – *one who computes* (1646).
- Computer – is a '*mechanical calculating machine*' (1897)
- Computer – is an *electronic machine/device* (1946) for storing and processing data.

4

What is Computer Science?

- It is *NOT* just about assembling computers or writing computer programs.
- It is *NOT* about the tools we use to carry out computation but it is about how we use such tools,
- It is *NOT* about *using* software (e.g. Excel, Word, power point, etc).
- It is *NOT* about expertise in computer games,
- it is *NOT* about writing content in websites, and

5

What is Computer Science?

- It is the **science of using computers** to solve problems – *George Washington University*
- It is the study of **computers and computational systems**, software and software systems including their **theory, design, development, and application** - *University of Maryland*
- Computer science is the study of computers and computational systems. It is a broad field which includes everything from the algorithms that make up software to how software interacts with hardware to how well software is developed and designed – *Michigan Tech University*

6

What is Computer Science?

Computer Science is about the -

- PRINCIPLES behind building the software packages,
- ALGORITHMS used in computer games, solving problems. It is an unambiguous, finite description in simple steps or actions
- TECHNOLOGY behind the Internet and
- ARCHITECTURE of computing devices.
- **Information** - knowledge represented in a form suitable for transmission, manipulation, etc.
- **Protocol** -- rules for exchanging information without problems

7

Computer Science:

- A discipline...
 - spans **theory** and **practice**.
 - **thinks** both in **abstract** and **concrete** terms.
 - involves understanding and design of computers and computational processes
- A science...
 - modelling and analyzing problems
 - Provides the methodology for learning and refinement
 - Designing and verifying solutions
 - Requires precision, creativity, and reasoning

8

Computer Science = Mathematics + Science + Engineering

Mathematics ---- the science of numbers, interrelations, and abstractions.

- Origin of computer science.
- Provides **reasons** and **logic**
- Essentially studies algorithms and principles of computing

Science --- Systematic knowledge or practice. Acquiring knowledge through the scientific method of natural phenomena (natural sciences) or human or social behavior (social sciences).

9

Computer Science = Mathematics + Science + Engineering

Engineering --- the applied science of acquiring and applying knowledge to design, or construct works for practical purposes.

- Provides techniques for building hardware and software
- Studies computer architecture – hardware & software

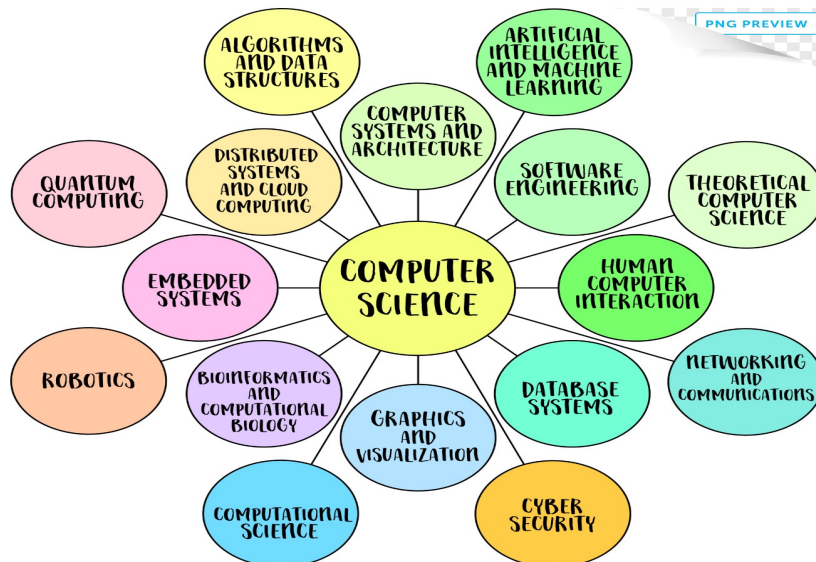
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Computer Science - In general...

- A central focus is on processes for handling and manipulating information
- Involves understanding and design of computers and computational processes
- Ranges from theoretical studies of algorithms to practical problems of implementation
- Concerned with *“the study of symbol-manipulating machines, with communication between man and machine and with the application of these machines”*

11

Areas of Computer Science



12

Theoretical Computer Science

1. Theory of Computation (CMSC 141, 142)

- explores the fundamental principles of computation, including topics like automata theory, formal languages, and computational complexity.
- It helps in understanding the capabilities and limitations of computers.

2. Algorithms and Data Structures (CMSC 10, 18, 28, 122)

- study algorithms (step-by-step procedures for solving problems) and data structures (ways of organizing and storing data) to design efficient and effective software solutions.
- Develops efficient methods for solving computational problems and organizing data

3. Programming Languages (CMSC 124, 129)

- CS involves writing, testing, and maintaining computer programs using various programming languages.
- Programmers and software developers are professionals who create applications, software systems, and algorithms.

4. Formal methods (CMSC 56, 57)

- Uses mathematical techniques to prove the correctness of software systems.

9/2/25

13

Applied Computer Science

1. Artificial Intelligence (AI) – CMSC 170

- It is a subset of computer science that deals with creating intelligent machines capable of mimicking human-like cognitive functions such as learning, problem-solving, and decision-making.
- Example - Chatgpt, Netflix recommender system

2. Machine Learning and Data Science

- CS plays a crucial role in the development of machine learning algorithms and data analysis techniques, enabling systems to learn from data and make predictions or decisions.

3. Computer Graphics and Visualization

- Computer scientists work on creating visual representations of data and 3D graphics for applications such as video games, virtual reality, and data visualization.

14

Key Aspects of Computer Science

4. Databases (CMSC 126, 127)

- design and management of databases, which are essential for storing, retrieving, and manipulating large sets of structured data efficiently.

5. Data Communication and Networking (CMSC 137)

- focuses on the design, implementation, and management of computer networks and the underlying hardware and software systems that make up modern computing infrastructure.

6. Security and Cybersecurity (CMSC 185)

- Focuses on development of strategies and tools to protect computer systems and networks from unauthorized access, data breaches, and cyberattacks.

7. Software Engineering (CMSC 128)

- covers software development methodologies, project management, and software quality assurance to ensure that software projects are completed on time and meet their intended goals

15

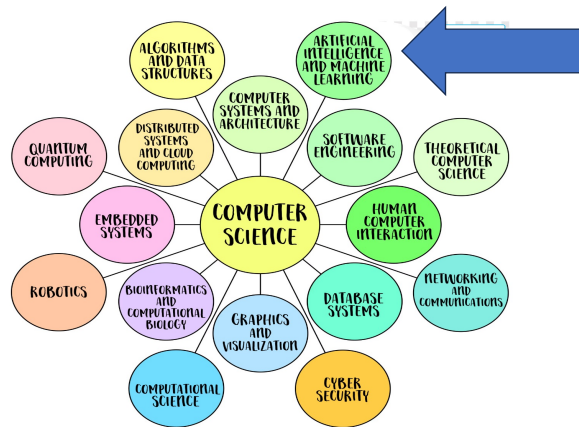
Key Aspects of Computer Science

8. Operating Systems (CMSC 125).

- software systems that manage computer hardware and provide a platform for running other software applications.
- concerned with the development and structure of complex programs which facilitate man-machine communications
- Operating Systems and Languages used for their development:
 - *Windows* - C, C++, C#, and .NET languages (e.g., VB.NET)
 - *Linux* - C and C++
 - *macOS* - Objective-C, Swift, C, and C++
 - *Unix* - Primarily C, with some assembly language.
 - *Android OS* - C and C++.
 - *iOS* - C, C++, and assembly language.
 - *RTOS (Real-Time Operating Systems)* - C and C++.

16

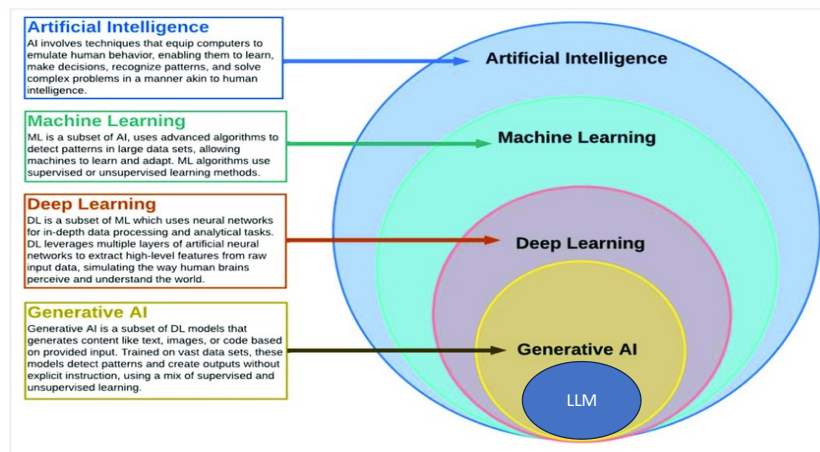
What is Artificial Intelligence (AI)?



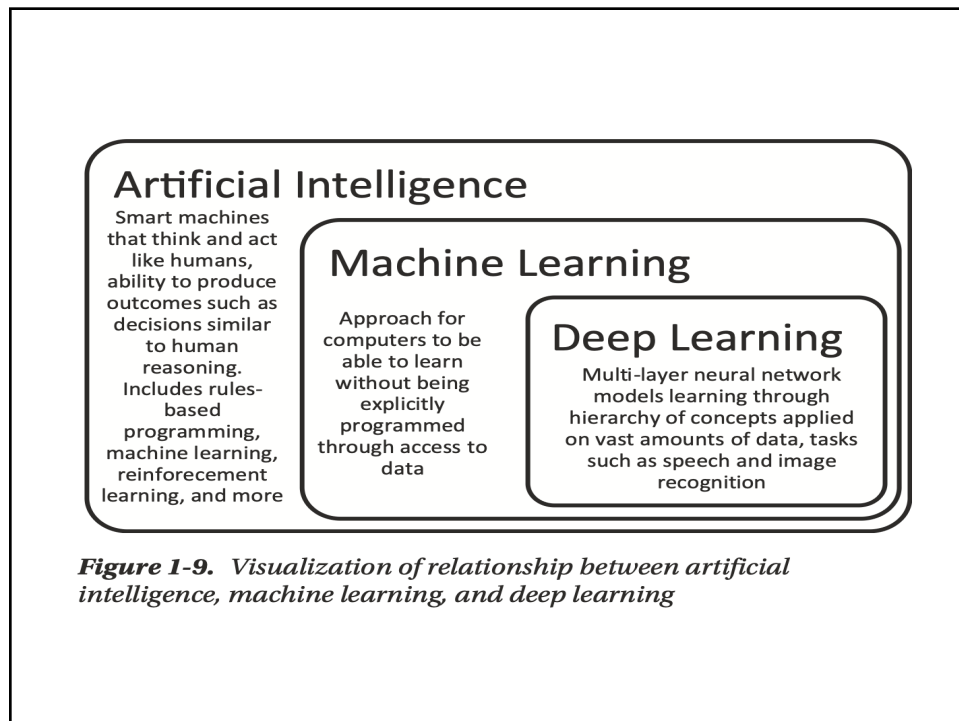
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17

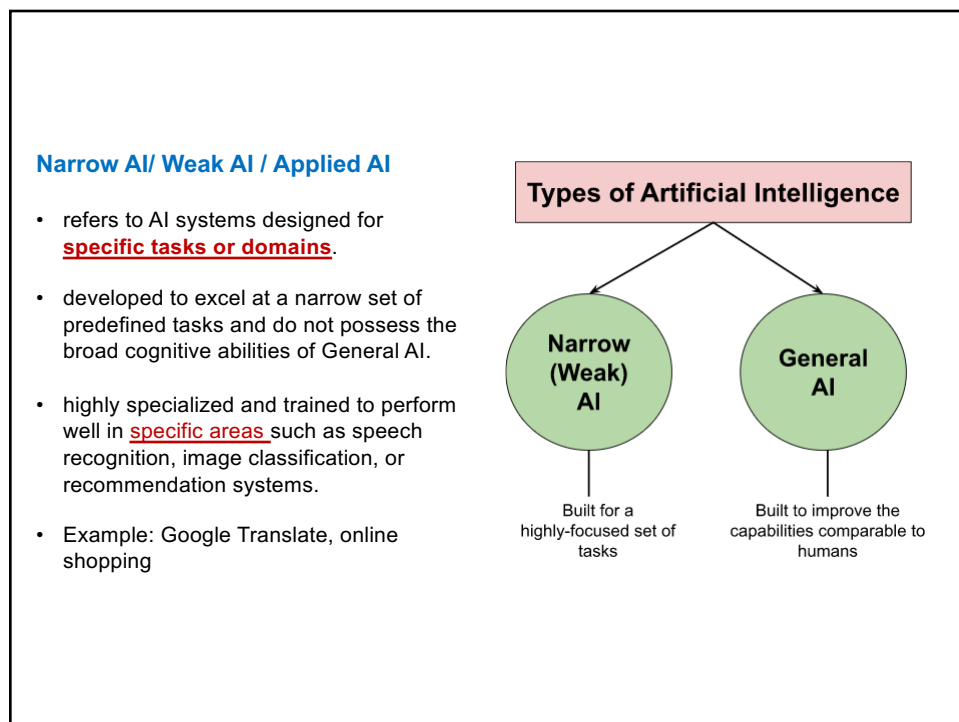
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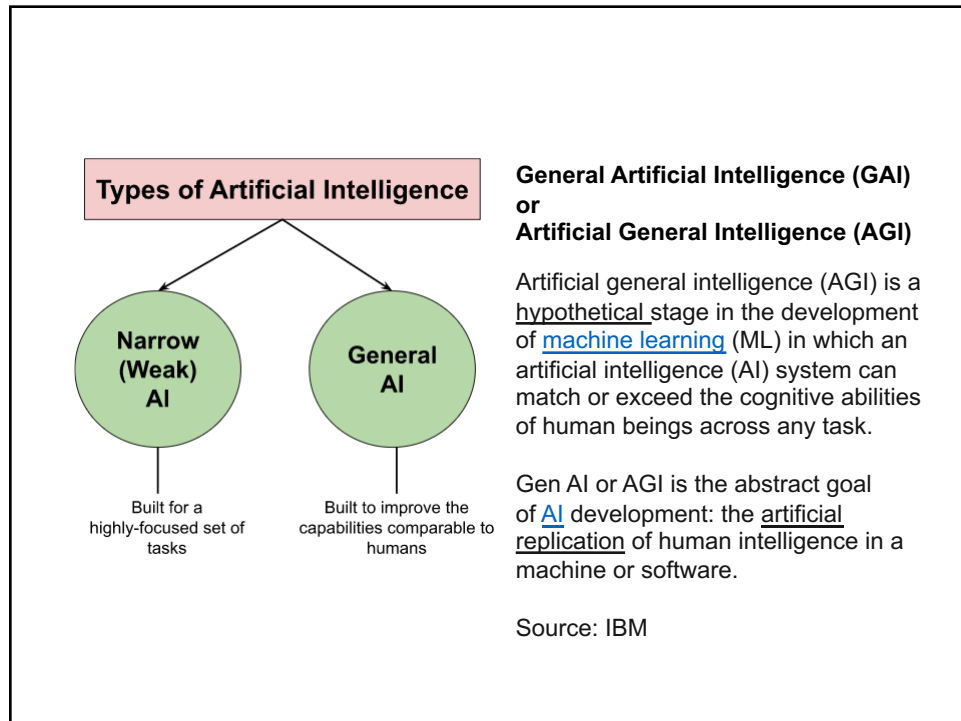
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19



20



21

Examples of Narrow AI		
AI Applications	Description	Examples
Language Translation Services	Facilitate real-time translation of text and speech between numerous languages.	Google Translate, ChatGPT
Virtual Assistants	AI-powered assistants that perform tasks like setting reminders, answering questions, and controlling smart home devices through voice commands.	Siri, Alexa
Facial Recognition Systems	Identify individuals by analyzing facial features, commonly used in security and authentication processes.	Face ID, DeepFace
Recommendation Systems	Use AI algorithms to suggest movies, shows, music, or products based on user preferences and behavior.	Netflix, Shopee, Lazada, Amazon, Shein
Autonomous Vehicles	Self-driving cars that interpret sensor data, navigate roads, and make driving decisions within specific scenarios.	Tesla Autopilot, Waymo
Medical Diagnosis Tools	AI applications that analyze medical data, such as imaging scans, to detect diseases like cancer, aiding healthcare professionals in diagnostics.	IBM Watson Health, Aidoc

22

General AI Models/Platform

1. GPT-4 by OpenAI

- early version of AGI or GAI
- Released in March 2023, GPT-4 is a large language model capable of generating human-like text across various domains.
- Early experiments indicated that GPT-4 could solve novel and difficult tasks spanning mathematics, coding, vision, medicine, law, and psychology without needing specialized prompting.
- Its performance in these areas was strikingly close to human-level

2. Gato by DeepMind

- Introduced in May 2022, Gato is a multimodal neural network designed to perform a range of complex tasks, including engaging in dialogue, playing video games, and controlling robotic arms for block stacking.
- Its ability to learn multiple different tasks simultaneously and switch between them without forgetting previous skills has been described as a step toward general-purpose AI.

23

General AI Models/Platform

3. OpenCog (<https://hyperon.opencog.org/>)

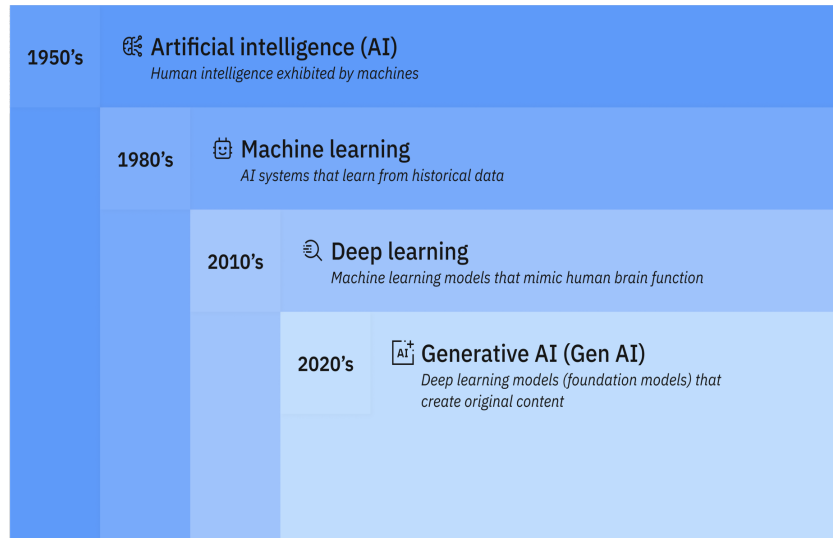
- an open-source project aimed at building a framework for artificial general intelligence.
- It defines a set of interacting components designed to give rise to human-equivalent AGI as an emergent phenomenon of the whole system.

4. OpenAI's o1-preview Model (<https://openai.com/index/introducing-openai-o1-preview/>)

- Released in 2024, this model demonstrates enhanced problem-solving abilities by allocating more computational resources to generate responses.
- it is still considered an advanced narrow AI system, lacking the comprehensive adaptability and understanding required for AGI.

24

History of AI



25

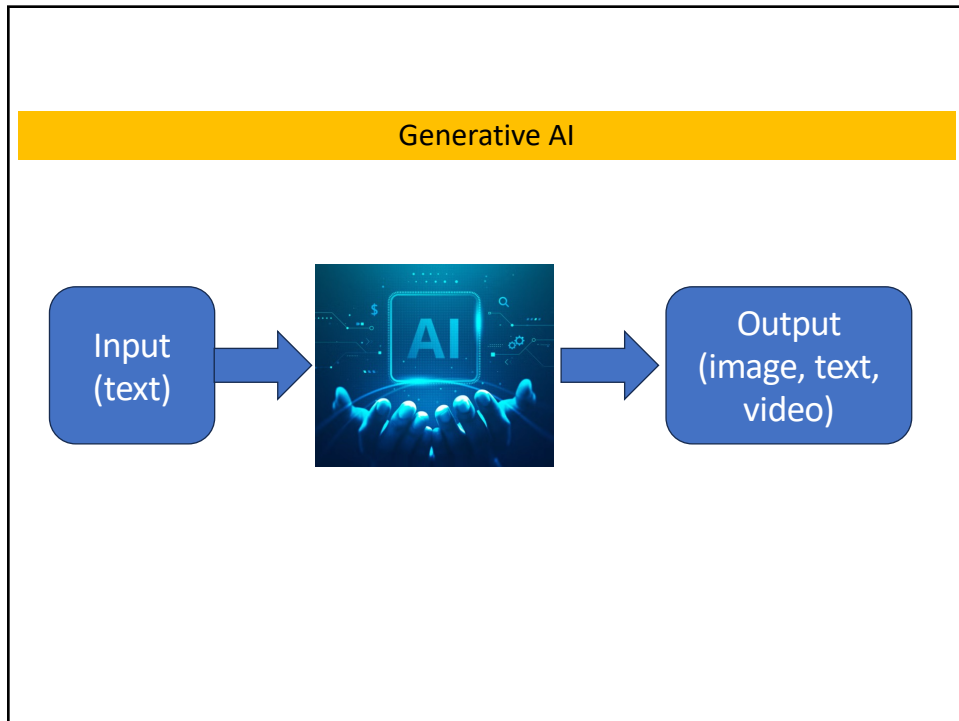
What is Generative AI or gen AI?

- Is an AI systems designed to **create** new content like text, images, music, or even code, based on the data they were trained on.
- Is a deep learning models that can create complex original content—such as long-form text, high-quality images, realistic video or audio and more—in response to a user's prompt or request.

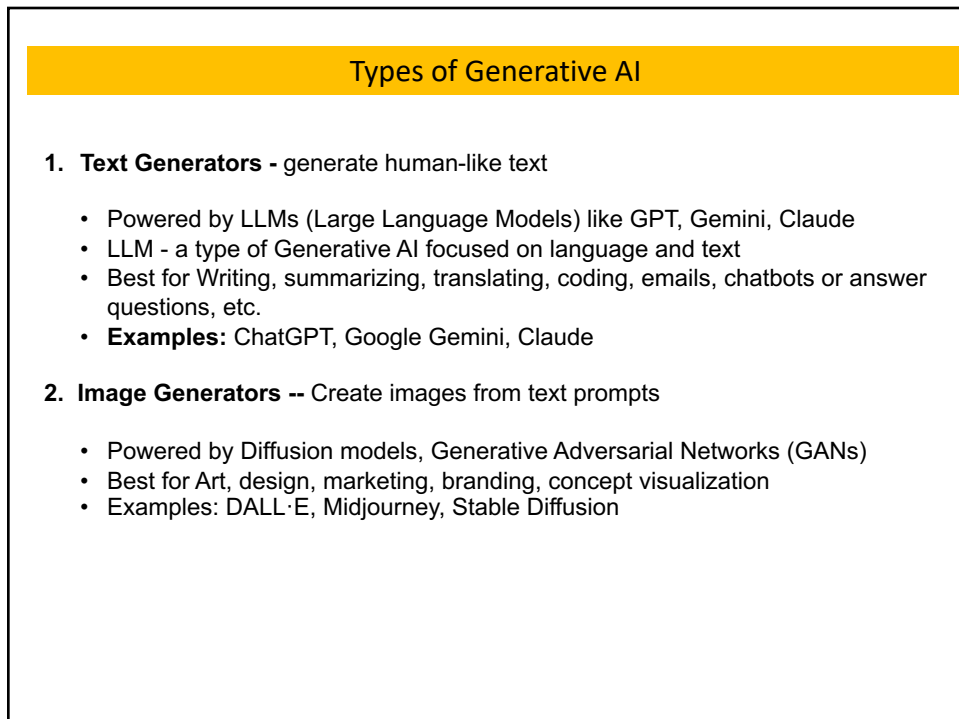
Main Types of Generative AI

1. Text Generators
2. Image generators
3. Video Generators
4. Audio Generators
5. Code Generators
6. 3D Model Generators
7. Multimodal Generators

26



27



28

Types of Generative AI

3. Video Generators - generate video clips from text, images, or other videos

- Best for Film, ads, social media content, virtual presenters
- Examples: Runway, Pika Labs, Synthesia, Sora (OpenAI)

4. Audio Generators - create music, sound effects, or speech from text or samples

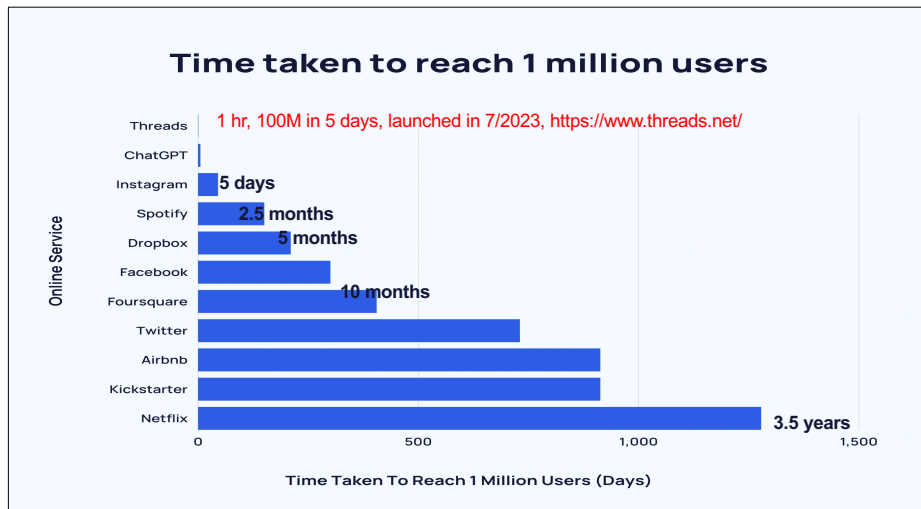
- Best for Podcasts, audiobooks, music production, dubbing
- Examples: ElevenLabs (speech), AIVA (music), Suno (music)

5. Code Generators - generate programming code based on text prompts

- Best for Software development, automation, debugging, scripting
- Examples: GitHub Copilot, Replit Ghostwriter, ChatGPT (with code capabilities)

29

Some Statistics on the Adoption of Text Generators (e.g. chatgpt)



30

Some Statistics on the Adoption of Text Generators (e.g. chatgpt)

According to Brad Lightcap, OpenAI's Chief Operating Officer, ChatGPT has 400 million weekly users. It plans to hit **1 billion** users by the end of 2025.

Month	Weekly Users
November 2022	1 million
January 2023	30 million
November 2023	100 million
December 2024	300 million
February 2025	400 million

Source: <https://explodingtopics.com/blog/chatgpt-users>

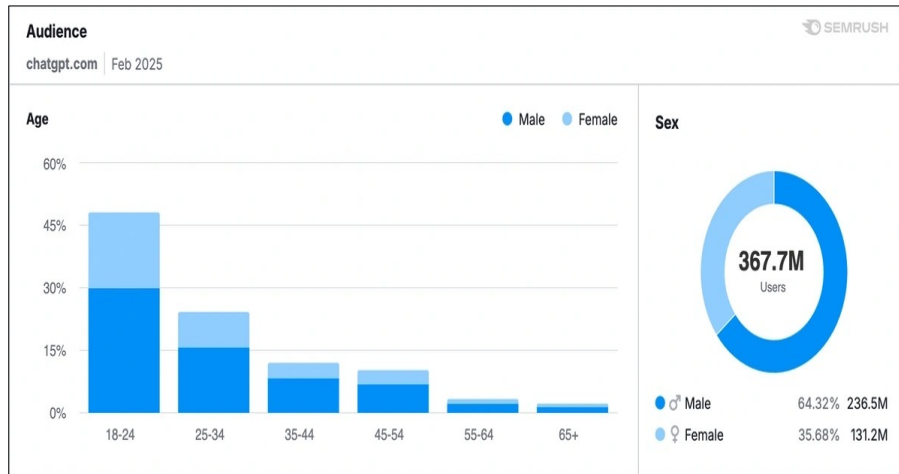
31

Chatgpt's Visit per month

Month	Number of Visits	Change Over Previous Month	Change Over Previous Month (%)
December 2022	264.7 million	-	-
January 2023	683.2 million	↑ 418.5 million	↑ 158.10%
February 2023	1.3 billion	↑ 616.8 million	↑ 90.28%
March 2023	1.9 billion	↑ 600 million	↑ 46.15%
April 2023	2.3 billion	↑ 400 million	↑ 21.05%
April 2024	2.8 billion	↑ 500 million	↑ 21.74%
November 2024	4.8 billion	↑ 2 billion	↑ 71.43%
December 2024	4.8 billion	-	-
January 2025	4.7 billion	↓ 0.1 billion	↓ 2.08%
February 2025	5.2 billion	↑ 0.5 billion	↑ 10.64%

32

Some Statistics on the Adoption of Text Generators (e.g. chatgpt)



Source: <https://explodingtopics.com/blog/chatgpt-users>

33

Chatgpt Awareness

Level of ChatGPT Awareness	Age Group			
	18 to 29	30 to 44	45 to 64	65+
I've used it myself to generate text	15%	17%	9%	5%
I've seen text generated for someone else, but haven't used it myself	48%	46%	27%	30%
I've never used or never seen anyone else use it	20%	31%	55%	59%
Not sure	17%	6%	8%	6%

34

Types of Generative AI		
Type	Output	Example Tools
Text	Essays, chats	ChatGPT, Gemini, Claude
Image	Art, photos	Midjourney, DALL·E
Video	Clips, animations	Runway, Sora
Audio	Music, voice	ElevenLabs, Suno
Code	Programming	GitHub Copilot, ChatGPT
3D Models	Objects, spaces	GET3D, DreamFusion
Multimodal	Mixed formats	GPT-4, Gemini

35

AI Models/Platforms for Text Generation			
AI Model	URL	Description	Use Cases
ChatGPT	https://chatgpt.com	Powerful language model for generating human-like text based on prompts.	Customer support, writing assistants, coding help, tutoring.
Gemini	https://gemini.google.com	Conversational AI integrated with Google Search for real-time information.	Information retrieval, web-based queries, real-time knowledge.
Claude	https://claude.ai	AI focused on safety and ethics, avoiding harm or bias in responses.	Safe AI applications, ethical decision-making, research assistance.
Bing Chat	https://www.bing.com/chat	Powered by GPT-4, integrates search engine and dialogue capabilities.	Search queries, content generation, research.
Jasper	https://www.jasper.ai	Content creation AI for marketing, blogs, and SEO optimization.	Content writing, copywriting, social media management, marketing.
Mistral AI	https://mistral.ai	Open-weight models for customized AI solutions and research.	Custom AI development, research, machine learning solutions.

36

AI Models/Platforms for Text Generation			
AI Model	URL	Description	Use Cases
LLaMA	https://www.meta.ai/	Open-source language model for text generation and analysis.	Natural language processing, research, AI development.
Cohere	https://cohere.ai	Platform for building natural language understanding models for businesses.	AI-driven customer service, business analytics, content generation.
YouChat	https://you.com	Conversational AI combined with a search engine for a streamlined experience.	Real-time answers, web-based queries, privacy-focused AI interaction.
Grok	https://x.com/i/grok	Social media-integrated AI for real-time conversations on X (formerly Twitter).	Social media interaction, personalized responses, content creation on X.
Replika	https://replika.ai	Emotionally intelligent chatbot for personal interaction and companionship.	Emotional support, personal assistant, companionship.
OpenAI Codex	https://openai.com/product/codex	AI model specialized for coding and software development tasks.	Software development, code generation, technical assistance.
IBM Watson	https://www.ibm.com/watson	AI suite for natural language processing, machine learning, and data analysis.	Business analytics, healthcare diagnostics, data-driven insights.
Hugging Face	https://huggingface.co	Open-source platform with pre-trained models for NLP, image recognition, etc.	AI development, machine learning research, custom model training.

37

AI Models/Platforms for Text Generation		
Platform	Best for	Pricing
DeepSeek	Coding, reasoning, and math	Free
Perplexity	Searching for statistics, links, and sources	\$20 per month
Claude	Natural language	\$20 per month
Google Gemini	Using AI connected to Google products	\$19.99 per month
Microsoft Copilot	AI-powered productivity assistant	\$20 per month
Meta AI	Integrating with social media and messaging apps	Free
Grok	Generating helpful and humorous responses with no limit	\$16 per month
ChatPDF	Summarizing and interacting with PDFs	\$20 per month
YouChat	AI-powered searches	\$20 per month
AnonChatGPT	Anonymous, login-free ChatGPT use	Free

38

AI Models/Platforms for Text Generation

Platform	Best for	Pricing
Twain	Outreach message optimization and coaching	Free
Jasper	Writing different kinds of copy and content	Starting at \$49 per month
Character.AI	Interacting with character-flavored chatbots	\$9.99 per month
IBM watsonx	Enterprise-level AI	Contact sales
Otter AI	Searching and summarizing meeting notes	Starting at \$16.99 per month
Auto-GPT	Creating automated processes	Starting at \$25.99
GitHub Copilot	Writing code more efficiently	Starting at \$3.67 per month
Writesonic	Fast writing, paraphrasing, and shortening	Starting at \$15 per month
Poe	Conversational AI assistant related to Quora	\$199 billed annually
Tome	AI-powered storytelling	\$16 per month
Amazon CodeWhisperer	Generating code based on natural language	\$19 per month

39

AI Models for Image Generation

Platform	Description	Example Use Case
Midjourney	An AI image generator known for producing high-quality, artistic images based on textual prompts.	Creating stylized artwork for creative projects.
DALL-E 3	OpenAI's advanced image generator capable of handling complex queries with various editing and customization options.	Generating detailed images from intricate textual descriptions.
Stable Diffusion	An open-source model that generates images from text prompts, offering flexibility and customization.	Developing unique images for design and artistic purposes.
Adobe Firefly	Adobe's AI image generator providing unique features like camera angle adjustments and a variety of customization options.	Enhancing images with specific artistic styles and compositions.
Canva's AI Image Generator	Canva's tool allows users to create AI-generated images and art using text prompts, integrating models like DALL·E by OpenAI and Imagen by Google Cloud.	Designing marketing materials with custom images.

40

Text Generation AI

Tools: gemini, perplexity, chatgpt, claude, Microsoft copilot

Tasks:

1. Writing (letter, email, poems, SPA, Songs, Project proposals, exam questionnaire)
2. Answer Questions/chatbot (e.g. What is Climate Change)
3. Summarizing documents
4. Summarizing (documents, word, minutes, pdf, videos, etc)
5. Translation
6. Program coding, Exam preparation
7. Document Preparation
8. Retrieval-Augmented Generation (RAG)

41

Prompt Engineering



The volume of soil that you can take depends on the tool that you use.

42

Prompt Engineering

Prompt 1: **Buy a drink.**



43

Prompt Engineering

Prompt 2: Buy a soft drink.



44

Prompt Engineering

Buy a bottle of soft drink from the sari-sari store



45

Prompt Engineering

Buy a 1-liter bottle of Coke from Aling Nena's sari-sari store.



46

Prompt Engineering

Go to Aling Nena's sari-sari store and buy a 1-liter bottle of Coke. If it's not available, get Pepsi. Use the ₱100 I gave you and bring back the change.



47

What is Prompt Engineering?

Prompt engineering -- is the process of designing and refining prompts to **effectively guide** generative AI models, such as chatbots or content generation tools, to produce desired outputs.

A prompt --- is a specific input or question given to an AI model to elicit a response. It can include instructions, context, or examples.

Goal: To optimize the AI's responses by crafting prompts that are clear, direct, and relevant.

The quality of the AI's response depends on the quality of your prompt.

48

Computer Science vs IT vs Computer Engineering

49

	BS CS	BS IT	BS Co Eng
Focus/ Purpose	- theoretical and scientific aspects of computing. - deals with algorithms, data structures, software development, programming languages, and the mathematical foundations of computing. - The goal of a CS program is to teach students how to design and create efficient and innovative software and systems.	- practical application of computing technology to solve real-world problems. - IT programs emphasize the management and use of technology resources, including hardware, software, networks, and databases, to support business operations and organizational goals.	- combines elements of electrical engineering and computer science. - It focuses on the design, development, and integration of computer systems, including both hardware and software components.
Core Courses	- CS programs typically include courses in algorithms, data structures, computer organization, software engineering, programming languages, theory of computation, and artificial intelligence. - It often has a strong emphasis on programming and problem-solving skills.	- database management, network administration, cybersecurity, system analysis and design, IT project management, and information systems. - Focuses on practical skills for managing and maintaining technology systems.	Digital logic design, comp architecture, microprocessors, Networks, micro controllers, signal processing, circuit analysis

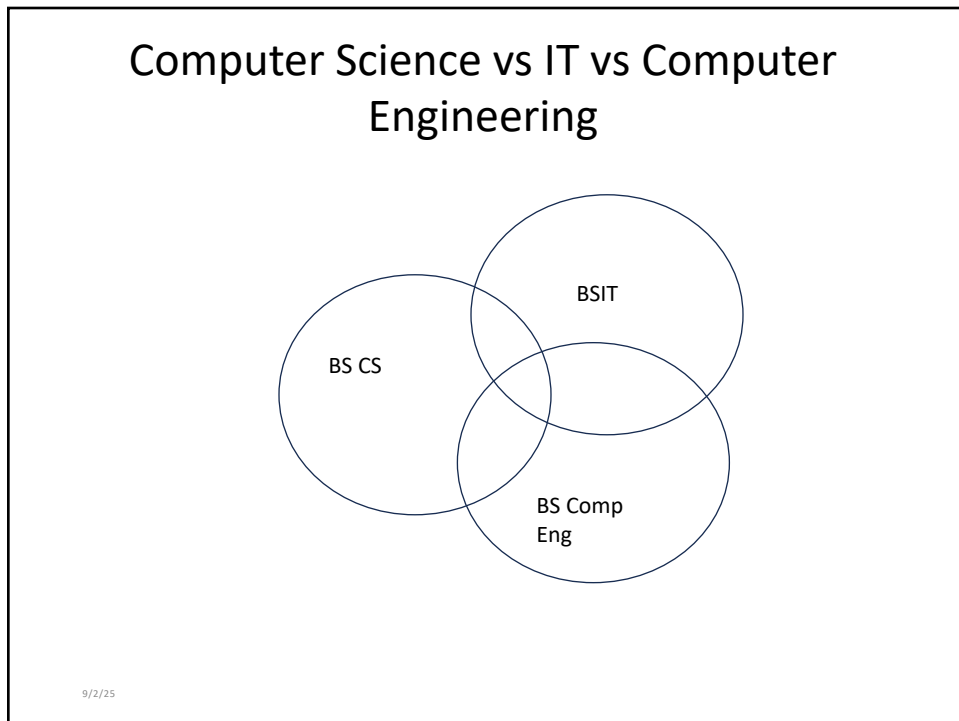
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50

	BS CS	BS IT	BS Co Eng
Career	- software developers, computer scientists, data scientists, algorithm designers, and researchers. - They can work in a wide range of industries, including technology, finance, healthcare, and academia.	- network administrators, database administrators, IT managers, cybersecurity specialists, system analysts, and IT consultants. - They are typically involved in the management and maintenance of IT infrastructure.	hardware design, embedded systems development, semiconductor manufacturing, robotics, and IoT (Internet of Things) industries.
Skills	- programming skills, algorithmic thinking, and problem-solving. - Graduates are skilled in software development, algorithm design, and computational theory.	- focus on skills related to the implementation, maintenance, and management of technology systems. - This includes network configuration, database management, cybersecurity, and system administration	possess a range of technical, analytical, and problem-solving skills that enable them to design, develop, and maintain computer systems and hardware.
Mathematics	require more advanced mathematics, including courses in calculus, discrete mathematics, and linear algebra	less emphasis on advanced mathematics and focus more on practical technical skills.	calculus, linear algebra, discrete math, differential equations, statistics & probabilities, numerical methods

9/2/25

51



52

Is Computer Science really for me?

53

Other Skills

1. Soft skills - such as teamwork, verbal and written communication, time management, problem solving, and flexibility
2. Personal attributes - such as risk tolerance, collegiality, patience, work ethic, identification of opportunity, sense of social responsibility, and appreciation for diversity, play a critical role in the workplace.

54

Reading Assignment:

1. <http://online.king.edu/information-technology/difference-between-a-computer-science-information-technology-degree/>
2. [Tips for Comp Sc Students](https://www.indeed.com/career-advice/career-development/computer-science-student-study-tips)
<https://www.indeed.com/career-advice/career-development/computer-science-student-study-tips>
3. How is Computer Science Different from IT?
<http://tinyurl.com/yusdmcv3>
4. Why Is Computer Science Important?
<http://tinyurl.com/3b8cfs29>

55

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2. <https://www.acm.org/education/CS2013-final-report.pdf>
3. <http://www.cs.bu.edu/AboutCS/WhatIsCS.pdf>
4. <http://online.king.edu/information-technology/difference-between-a-computer-science-information-technology-degree/>
5. <https://www.indeed.com/career-advice/career-development/computer-science-student-study-tips>

56

Thank you!